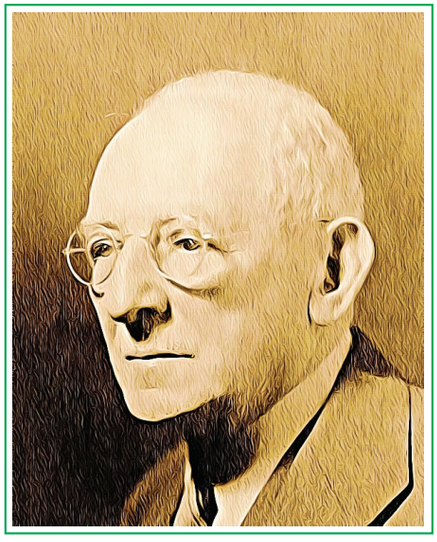
Unit 1 BASIC CONCEPTS OF BIO CHEMISTRY AND CELL BIOLOGY

DIDQH7



Carl Alexander Neuberg

Carl Alexander Neuberg, often referred to as the father of modern biochemistry is a German-Jewish pioneer in biochemistry. He gained international recognition through his elucidation of the biochemical reactions of alcoholic fermentation in which he discovered a number of different enzymes such as carboxylase and of intermediates such as fructose-6-phosphate. This understanding had become crucial as to how metabolic pathways would be investigated by later researchers.

Learning Objectives

After studying this unit, the student will be able to

- Describe the cell
- Classify cells into different types
- Differentiate between prokaryotic and eukaryotic cells
- Explain the shapes and sizes of different cells
- Describe the functions of different cellular organelles
- Calculate pH from hydrogen ion concentration
- Apply Henderson-Hasselbalch equation to calculate the pH of a buffer
- Describe the intracellular and extracellular fluids
- State the components of blood and lymph
- Identify the pH of different body fluids
- Explain the role of buffers in regulation of pH
- Describe the importance of lungs and kidneys for acid-base regulation
- Measure the pH of different solutions using pH paper and pH meter

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